# WEATHER quick reference

**Version 1.0** — reasoning from your own observations, no network.

## Commands

| | |
|---|---|
| `weather` | the menu |
| `weather what` | what your log says is coming, with the reasoning |
| `weather learn [key]` | a lesson; no key lists them |
| `weather chart` | reason over a 500 mb radiofax chart |
| `weather forecast` | yours first, then mine, then both scored |
| `weather score` | you, the rules and persistence |
| `weather lon <deg>` | your longitude, east positive |
| `weather day\|night\|plain` | colour mode |
| `weather about` / `version` | |

## The pressure, in 3 hours

| | |
|---|---|
| fall of 3.0 hPa or more | a serious fall: a deepening system and a rising wind |
| fall of 1.5 to 3.0 | a definite fall — something is coming |
| within 1.5 either way | near steady |
| rise of 1.5 or more | the gradient is easing, or a ridge building |

**The rate matters more than the level.** 1000 steady is a quiet day; 1015 falling fast is not. The figures are corrected for the atmospheric tide first, which needs your longitude.

## Buys Ballot

Back to the wind: **northern hemisphere, the low is on your left**; southern, on your right.

**Backing wind, falling glass — the centre is to your north and closing.** That is the one to take seriously.

## Cloud base and fog

| | |
|---|---|
| Cloud base | the temperature–dew point spread closes at **2.5 °C per 1000 ft** |
| | (a parcel cools at 3, its dew point falls at 0.5; the difference is what counts) |
| Fog | watch the margin between **dew point and sea temperature** |

## The warm front, in cloud

**cirrus → cirrostratus → altostratus → nimbostratus and rain.** Logged by level, it is readable in advance. Logged as “overcast”, it is not.

## The 500 mb chart — five numbers off the fax

bearing to the **564 line** · its distance · the line’s orientation · the 500 mb wind there · hemisphere.

**Equatorward of the 564 line is where you want to be.**

## The forecast loop

1. **Yours is written first.** Nothing of the machine’s until it is.
2. Then the **rules** forecaster and **persistence**.
3. All three scored against the verifying observation: half a point per knot, one per millibar, a tenth per degree.

**Persistence is the benchmark.** A forecaster who cannot beat it is not yet adding anything.

## What it will not do

1. It cannot forecast: no models, no GRIBs, no charts over the wire.
2. It cannot see what has not announced itself in your observations.
3. It is only as good as the instruments and the care behind the log.